

Q1

	ARCHER2	Bede
Nodes	5,860	32
Processor	2× AMD EPYCTM 7742, 2.25 GHz, 64-core	2x, IBM Power9, 2.7 GHz, 16-core
Cores per node	128 (2 x 64-core processor)	32 (2x 16-core processor)
Memory per node	256 GiB (standard memory), 512 GiB (high memory)	512GB DDR4 RAM
Interconnect	HPE Cray Slingshot, 2× 100 Gbps bi-directional per node	NVLink 2.0
GPU per node	Only 4 nodes with 4x AMD Instinct MI210	4x NVIDIA Tesla V100 32GB NVLink 2.0 GPUs (2 per CPU)

It is clear from the table that ARCHER2 is focused on a CPU architecture, with 750,080 cores in total. ARCHER2 has an estimated peak performance of 28 PFLOPs. However, Bede is focused on a GPU architecture, boasting many more high powered GPUs than ARCHER2.

- ARCHER2 is designed for general purpose high performance computing, running programs such as CASTEP and HadGEM3. The use of 64 core processors and a large amount of nodes means that ARCHER2 offers extreme benefits from parallel programming. On the other hand, Bede is designed for more specialised applications that need GPU resources such as machine learning, a more modern demand.
- For large parallel programs, the communication between different pieces of hardware is extremely important. ARCHER2 has a 100 Gbps interconnect which is quite fast and does help minimise communication time. However Bede's NVLink 2.0 offers a 25 Gigatransfers per second, which is an estimated 1600 Gbps. This much higher interconnect is needed for the high memory usage that GPU programs typically need.

- ARCHER2 has a focus on scalability that programs from a variety of disciplines can be ran on it, making it a central computational resource for the wider country. However, Bede has more of a focus on specialised studies such as AI, machine learning and neural networks and the decision to focus on GPUs gives researchers a great computational resource for their own studies.

For GROMACs, both supercomputers offer their own benefits to being used. However I think that Bede would be more suitable for a GROMACs simulation as with all molecular dynamic simulations, there is a heavy cost of computation on solving Newton's equations. For a large number of particles, there will be a lot of computation necessary, to which GPU acceleration excels at. Another benefit to using Bede is the high memory interconnect, which means the communication between the heavy loaded GPU and CPU is still small compared to the size of the simulation. ARCHER2 does have a large amount of compute nodes, however due to the memory interconnect and the overhead associated with running the nodes, I expect that ARCHER2 would perform a lot worse for large simulations compared to Bede.